



COURSE DESCRIPTION FORM

INSTITUTION National University of Computer and Emerging Sciences (NUCES-FAST)

PROGRAM (S) TO BE BS(CS)

EVALUATED

A. Course Description

B.

(Fill out the following table for each course in your computer science curriculum. A filled out form should not be more than 2-3 pages.)

Course Code	EE(1005)																																																																																	
Course Title	Digital Logic Design (DLD)																																																																																	
Credit Hours	3+1																																																																																	
Prerequisites by Course(s) and Topics																																																																																		
Assessment Instruments with Weights (homework, quizzes, midterms, final, programming assignments, lab work, etc.)	Assessment Tools (AT) & Criteria <table border="1"> <thead> <tr> <th>AT</th><th>Description</th><th>Weight</th></tr> </thead> <tbody> <tr> <td>1</td><td>Mid Term I and II</td><td>30</td></tr> <tr> <td>2</td><td>Assignments</td><td>10</td></tr> <tr> <td>3</td><td>Quizzes</td><td>10</td></tr> <tr> <td>4</td><td>Final Examination</td><td>50</td></tr> </tbody> </table> AT to CLO Mapping <table border="1"> <thead> <tr> <th></th><th>AT 1</th><th>AT 2</th><th>AT 3</th><th>AT 4</th><th>Weight</th></tr> </thead> <tbody> <tr> <td>CLO 1</td><td>2</td><td></td><td></td><td>2</td><td>4</td></tr> <tr> <td>CLO 2</td><td>13</td><td>3</td><td>4</td><td>8</td><td>28</td></tr> <tr> <td>CLO 3</td><td>15</td><td>4</td><td>3</td><td>10</td><td>32</td></tr> <tr> <td>CLO 4</td><td></td><td>3</td><td>3</td><td>30</td><td>36</td></tr> </tbody> </table> AT to PLO Mapping <table border="1"> <thead> <tr> <th></th><th>AT1</th><th>AT 2</th><th>AT 3</th><th>AT 4</th><th>Weight</th></tr> </thead> <tbody> <tr> <td>PLO 1</td><td></td><td></td><td></td><td></td><td></td></tr> <tr> <td>PLO 2</td><td>15</td><td>3</td><td>4</td><td>10</td><td>32</td></tr> <tr> <td>PLO 3</td><td>15</td><td>7</td><td>6</td><td>40</td><td>68</td></tr> <tr> <td>PLO 4</td><td></td><td></td><td></td><td></td><td></td></tr> <tr> <td>PLO 5</td><td></td><td></td><td></td><td></td><td></td></tr> </tbody> </table>	AT	Description	Weight	1	Mid Term I and II	30	2	Assignments	10	3	Quizzes	10	4	Final Examination	50		AT 1	AT 2	AT 3	AT 4	Weight	CLO 1	2			2	4	CLO 2	13	3	4	8	28	CLO 3	15	4	3	10	32	CLO 4		3	3	30	36		AT1	AT 2	AT 3	AT 4	Weight	PLO 1						PLO 2	15	3	4	10	32	PLO 3	15	7	6	40	68	PLO 4						PLO 5					
AT	Description	Weight																																																																																
1	Mid Term I and II	30																																																																																
2	Assignments	10																																																																																
3	Quizzes	10																																																																																
4	Final Examination	50																																																																																
	AT 1	AT 2	AT 3	AT 4	Weight																																																																													
CLO 1	2			2	4																																																																													
CLO 2	13	3	4	8	28																																																																													
CLO 3	15	4	3	10	32																																																																													
CLO 4		3	3	30	36																																																																													
	AT1	AT 2	AT 3	AT 4	Weight																																																																													
PLO 1																																																																																		
PLO 2	15	3	4	10	32																																																																													
PLO 3	15	7	6	40	68																																																																													
PLO 4																																																																																		
PLO 5																																																																																		

	PLO 6						
	PLO 7						
	PLO 8						
	PLO 9						
	PLO 10						
	PLO 11						
	PLO 12						
Course Coordinator	Shahar Bano						
URL (if any)							
Current Catalog Description	The goal of this course is to introduce concepts & tools for the design of digital electronic circuits using sequential and combinational logic to the freshmen computer science students.						
Textbook (or Laboratory Manual for Laboratory Courses)	Digital Fundamentals , 11 th Edition, Floyd and Jain						
Reference Material	1. Digital Systems Principles and Applications 8th Ed, Tocci, Widmer and Moss 2. Digital Design by Moris Mano						

Course Goals	Course Learning Outcomes (CLO)				
	No.	Course Learning Outcomes (CLO)	Domain	Taxonomy Level	PLO
	1	Identify and explain fundamental concepts of digital logic design including basic and universal gates, number systems, binary coded system, basic components of combinational and sequence circuits.	Cognitive	2	2
	2	Demonstrate the acquired knowledge to apply techniques related to the design and analysis of digital electronics circuits, including Boolean Algebra and Multi-variable Karnaugh map methods.	Cognitive	3	2
	3	Analyze small –scale combinational digital circuits.	Cognitive	3	3
	4	Design small-scale combinational and synchronous sequential digital circuit using Boolean Algebra and K-map.	Cognitive	3	3
Relevant Program Learning Outcomes (PLOs):					
PLO 1	Computing Knowledge	Apply knowledge of mathematics, natural sciences, computing fundamentals, and a computing specialization to the solution of complex computing problems.			
PLO 2	Problem Analysis	Identify, formulate, research literature, and analyse complex computing problems, reaching substantiated conclusions using first principles of mathematics, natural sciences, and computing sciences.			
PLO 3	Design/Develop Solutions	Design solutions for complex computing problems and design systems, components, and processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.			
PLO 4	Investigation & Experimentation	Conduct investigation of complex computing problems using research based knowledge and research based methods			
PLO 5	Modern Tool Usage	Create, select, and apply appropriate techniques, resources and modern computing tools, including prediction and modelling for complex computing problems.			
PLO 6	Society Responsibility	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal, and cultural issues relevant to context of complex computing problems.			
PLO 7	Environment and Sustainability	Understand and evaluate sustainability and impact of professional computing work in the solution of complex computing problems			

	PLO 8	Ethics	Apply ethical principles and commit to professional ethics and responsibilities and norms of computing practice.
	PLO 9	Individual and Team Work	Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.
	PLO 10	Communication	Communicate effectively on complex computing activities with the computing community and with society at large.
	PLO 11	Project Mgmt and Finance	Demonstrate knowledge and understanding of management principles and economic decision making and apply these to one's own work as a member or a team.
	PLO 12	Life Long Learning	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological changes.

Relation between CLOs and PLOs (CLO: Course Learning Outcome, PLOs: Program Learning Outcomes)														
		PLOs												
		1	2	3	4	5	6	7	8	9	10	11	12	
CLOs	1		✓											
	2		✓											
	3			✓										
	4			✓										

Topics Covered in the Course, with Number of Lectures on Each Topic (assume 15-week instruction and one-hour lectures)	Week No.	Course Contents/Topics	Chapter	CLOs	Tools
	1	Decimal Numbers, Binary Numbers, Decimal-to-Binary Conversion, Binary Arithmetic, Complements of Binary Numbers, Signed Numbers, Arithmetic Operations with Signed Numbers, Hexadecimal Numbers, Octal Numbers , Binary Coded Decimal (BCD), Digital Codes (Gray code with conversion)	2	1	A1, M1, F
	2	Logic gates, The Inverter, The AND Gate, The OR Gate, The NAND Gate, The NOR Gate, The Exclusive-OR and Exclusive- NOR Gates	3	2	A2,Q1,M1,F

	3	Boolean Operations and Expressions, Laws and Rules of Boolean Algebra, DeMorgan's Theorems,	4	2	A2,M1,F
	4	Boolean Analysis of Logic Circuits, Logic Simplification Using Boolean Algebra, Standard Forms of Boolean Expressions, Boolean Expressions and Truth Tables.	4	2	A2,M1,F
		The Karnaugh Map, Karnaugh Map SOP Minimization, Karnaugh Map POS Minimization	4	2	A2,M1, F
	6	Mid-Term I			
	7	Basic Combinational Logic Circuits, The Universal Property of NAND and NOR gates, Pulse Waveform Operation	5	3	A3,M2, F
	8	Half and Full Adders, Parallel Binary Adders,	6	3	A3,Q2,M2 ,F
	9	Comparators, Decoders, Encoders,	6	3	A3,M2,F
	10	Code Converters, Multiplexers, Demultiplexers	6	3	A3,M2, F
	11	Mid-Term II			
		Latches			
	12		7	4	,F

	13	Flip-Flop	7	4	Q3, F
	14	Asynchronous Counters, Synchronous Counters, Design of Synchronous Counters	9	4	Q3, F
	15	Shift Register Operations, Types of Shift Register Data I/Os,	8	4	F
	16	Revision			
		Final Examination			
Laboratory Projects/Experiments Done in the Course					
Programming Assignments Done in the Course					
Class Time Spent on (in credit hours)	Theory	Problem Analysis	Solution Design	Social and I	
	30	10	5		
Oral and Written Communications	Every student is required to submit at least __1__ written report of typically __2__ pages and to make __1__ oral presentation typically __10__ minute's duration. Include only material that is graded for grammar, spelling, style, and so forth, as well as technical content, completeness, and accuracy.				

Instructor Name: Sana Naz

Instructor Signature

Date 19-01-26